

BACHELOR THESIS
ARTIFICIAL INTELLIGENCE

Radboud University



**Decision Support Systems for Radiology:
Ethical Safeguards in AI-Assisted
Diagnosis**

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Abstract

This thesis explores the ethical implementation of AI in medicine, with a particular focus on decision support systems (DSS) in radiology. Using the hypothetical DSS OrthoGuide, which assists in the classification of supracondylar humerus fractures, this thesis examines both the clinical benefits and the broader societal risks that accompany such technology. These systems can strengthen diagnostic consistency and ease the workload of medical experts, yet important concerns remain regarding data bias, privacy, and the possibility of reduced human oversight. To address these issues, a detailed risk assessment will be carried out, and the findings will be connected to the requirements of the EU AI Act. This will be followed by a design recommendation report that considers both technical and institutional safeguards needed to support responsible use. The aim of this study is to propose an approach to the integration of OrthoGuide that is accurate and at the same time responsible, ensuring that the tool supports clinical practice without compromising ethical standards. In doing so, the thesis seeks to bridge the gap between technical innovation and clinical utility, and to contribute to a more thoughtful and informed adoption of AI in medical settings.

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